

IE 5311-001 (D01)
Principles of Optimization
Fall 2026

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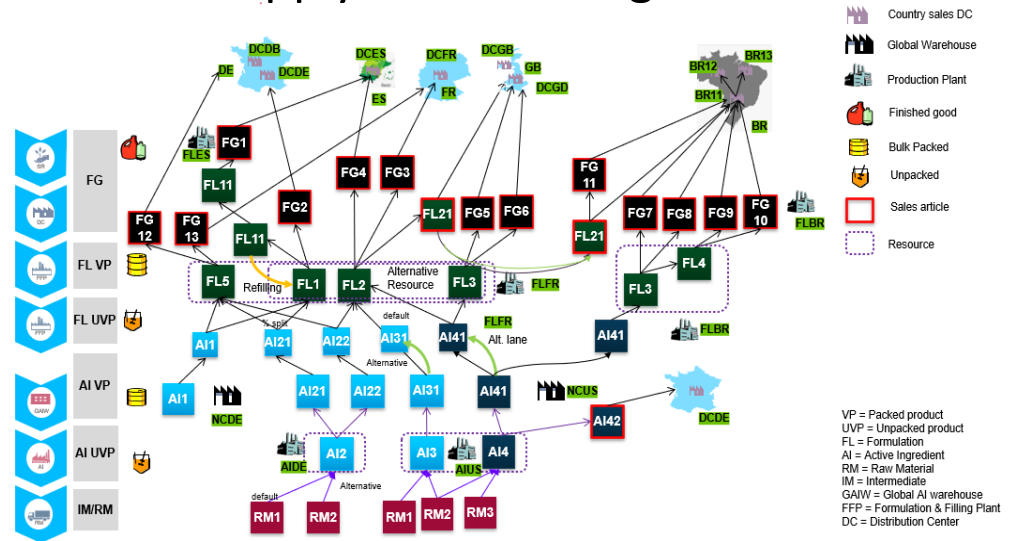
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Optimization is Everywhere 1

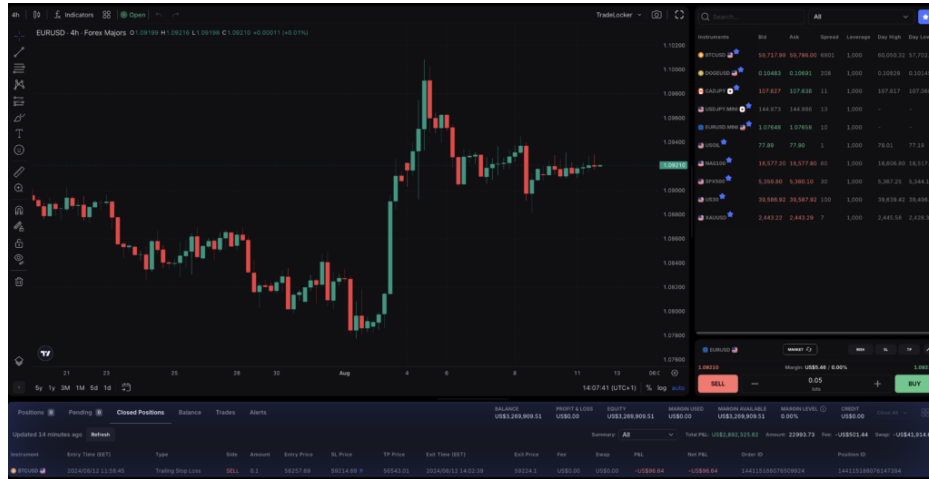


Multi-Armed Bandits

Supply Chain Management



Optimization is Everywhere 2



Portfolio Management

Metal Recycling

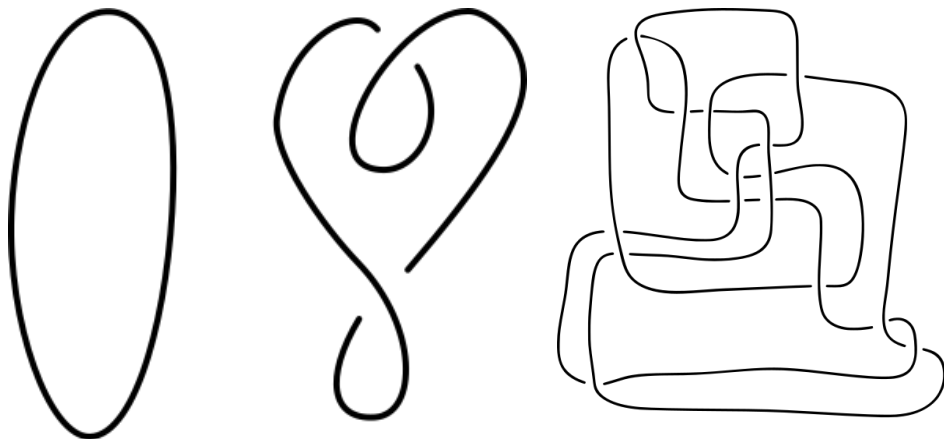


Optimization is Everywhere 3

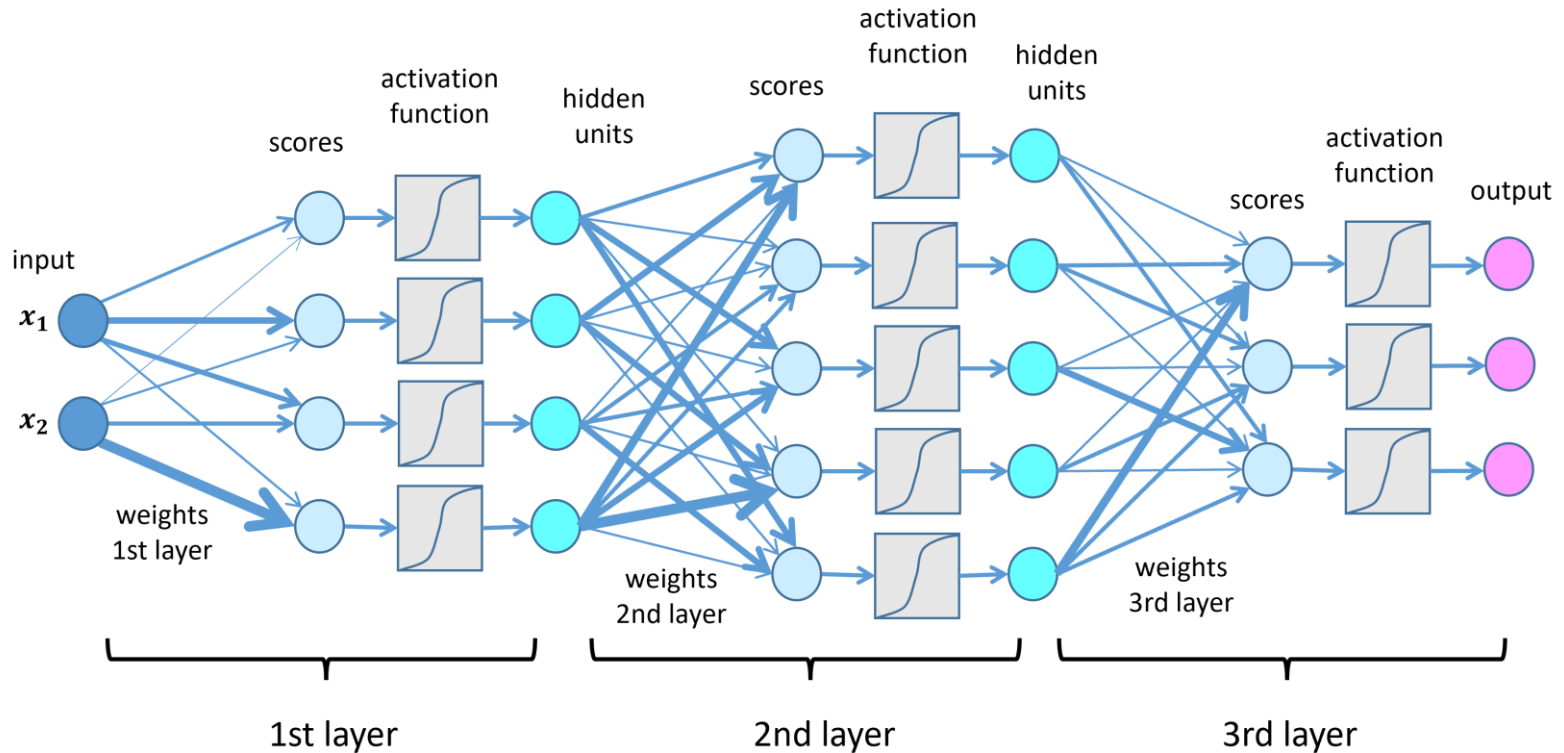


Sphere packing (Kepler conjecture)

Unknot detection



Optimization is Everywhere 4



Under the Hook

- General formulation
- From problem to formulation, from formulation to solution (40%)
- Understand the methods deeply -> improve / develop methods when needed (60%)



Preliminaries

- Linear algebra (in the language of vector, basis, linear maps, kernel, range, ...)
- Math proofs

Two self-check problems

- Consider the vector space of degree-2 polynomials with real coefficient, given two bases: $\mathcal{B}_1 = \{1, x, x^2\}$, $\mathcal{B}_2 = \{1 + x, x + x^2, 1 + x^2\}$, for the polynomial

$$p(x) = 3 + 5x + 4x^2$$

find its coefficients under $\mathcal{B}_1$ and $\mathcal{B}_2$, respectively.

- Given any matrix A , prove that the $\ker(A^\top A) = \ker(A)$.



Overview: The Breadth of Optimization 1

- General Form
- Constrained vs unconstrained
- Variable type: continuous, discrete, mixed
- Variable size: compact size, huge but finite, infinite dimensional



Overview: The Breadth of Optimization 2

- General Form
- Objective/Constraint type: linear, convex, nonlinear (nonconvex)
- Constraint size: compact size, huge but finite, semi-infinite
- Objective size: single objective, multi-objective



Overview: The Breadth of Optimization 3

- General Form
- Parameters: deterministic vs stochastic
 - Expectation optimization
 - Worst-case optimization
 - Chance-constrained optimization
- Decision-stage: one-stage, multi-stage
- Number of players: single, two-player, multiplayer
 - Acting simultaneously or sequentially



Overview: The Depth of Optimization

- Practice:
 - How to model (multiple ways)
 - How to reformulate (which one is better)
 - How to solve
- Theory:
 - Basic representations (finite / infinite-dim vector spaces)
 - Study of feasible regions (polyhedron theory, convex analysis)
 - Study of reformulations (duality theory)
 - Study of algorithm performance (complexity theory, convergence theory)
 - Network optimization (graph / matroid theory)
- Algorithm design:
 - Gradient methods (first-order, higher-order)
 - Simplex / interior point / barrier methods
 - GPU-enhanced algorithms / Quantum optimization



Course Outline

- Part 1: Classic Problems and Formulations
- Part 2: Linear Programming
- Part 3: Conic Programming

Exploration Credit: Statements or phrases marked with ** indicate exploration topics. You may choose to study these topics independently and write a concise review. Each successful exploration is worth **1 extra course point**, up to a maximum of **10 points** total.



Part I: Classic Problems and Formulations



Problem 1: Diet Problem (Stigler Diet)



History

Original version (1945):

For a moderately active man weighing 154 pounds, how much of each of 77 foods should be eaten on a daily basis so that the man's intake of nine nutrients will be at least equal to the recommended dietary allowances (RDAs) suggested by the National Research Council in 1943, with the cost of the diet being minimal?

Simplified version:

- Food: tomatoes, potatoes, beef, chicken, noodle, rice,...
- Nutrition: proteins, fats, carbs
- Each food has a price per unit, and each nutrition has a recommended level bounds
- What is the objective?

Heuristics: eliminated 62 foods, with value \$39.93



The Birth of Linear Programming

Original version (1945):

For a moderately active man weighing 154 pounds, how much of each of 77 foods should be eaten on a daily basis so that the man's intake of nine nutrients will be at least equal to the recommended dietary allowances (RDAs) suggested by the National Research Council in 1943, with the cost of the diet being minimal?

Who solved it (optimally): George Dantzig

When: 7 years later

Method: Linear Programming, Simplex Method

Optimal value: \$39.69

Rigorously formulate the diet problem as best as you can!



Mathematical Formulation

Simplified version:

- Food: tomatoes, potatoes, beef, chicken, noodle, rice,...
- Nutrition: proteins, fats, carbs
- Each food has a price per unit, and each nutrition has a recommended level bounds
- What is the objective?



Formulation Classification

Formulation:

- Constrained
- Variables: continuous, compact size
- Constraints: linear, compact size
- Objective: linear, single
- Parameter: deterministic
- Decision stage: one-stage
- # players: single



Basic Formulation Objects

- Linear function
- Hyperplane
- Half-space
- Intersections



Problem 2: Shortest Path



Basic Terminology in Graph Theory 1

- (Undirected) graph
- Neighborhood & Degree
- Cut



Basic Terminology in Graph Theory 2

- Walk
- Trail
- Path
- Cycle



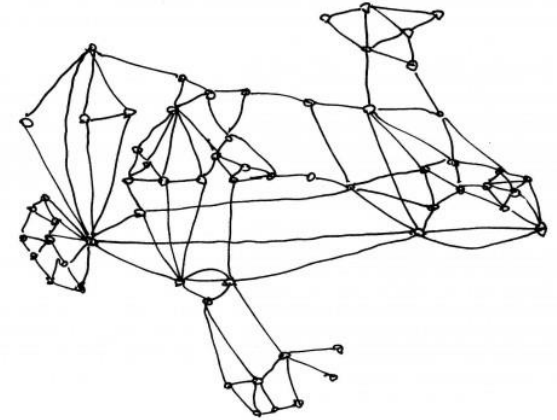
Basic Terminology in Graph Theory 3

- Directed graph
- In-neighborhood & In-degree
- Out-neighborhood & Out-degree
- In-cut & Out-cut



Shortest Path (Transportation/Logistics)

An emergency has occurred in region t ; the central policy department at location s must dispatch a team to reach the site as quickly as possible.



Is This Formulation Exact?



A Special Property of SP Formulation

An emergency has occurred in region t ; the central policy department at location s must dispatch a team to reach the site as quickly as possible.

** Total unimodularity explains why this relaxation is exact. Explore what total unimodularity means and why it guarantees an exact relaxation for this formulation.



SP Formulation Classification & How to Solve

An emergency has occurred in region t ; the central policy department at location s must dispatch a team to reach the site as quickly as possible.

- Constrained
- Variables: binary, compact size
- Constraints: linear, compact size
- Objective: linear, single
- Parameter: deterministic
- Decision stage: one-stage
- # players: single



Problem 3: Traveling Salesman Problem



Problem Setting (Dantzig–Fulkerson–Johnson Formulation)

In a complete undirected graph, a team must visit a set of locations and return to its starting point while minimizing the total travel distance.



DFJ Formulation

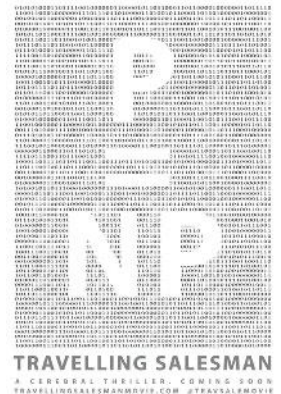
In a complete undirected graph, a team must visit a set of locations and return to its starting point while minimizing the total travel distance.

There is another Miller–Tucker–Zemlin (MTZ) compact formulation for TSP. They describe the same feasible solution space, but one is “tighter” than the other.



TSP: Fun Facts

- One of the Most Famous Discrete Optimization Problems
- Books: <https://www.amazon.com/Pursuit-Traveling-Salesman-Mathematics-Computation/dp/0691152705>
- Movie: https://www.imdb.com/title/tt1801123/?ref_=mv_close
- TSP art: <https://www.math.uwaterloo.ca/tsp/data/art/index.html>
- Exact solver Concorde: <https://www.math.uwaterloo.ca/tsp/concorde/index.html>
- Relation to complexity theory (computer science)



DFJ Formulation Classification

A team must visit a set of locations and return to its starting point while minimizing the total travel distance.

- Constrained
- Variables: binary, compact size
- Constraints: linear, huge size
- Objective: linear, single
- Parameter: deterministic
- Decision stage: one-stage
- # players: single



How to Solve DFJ Formulation?

Master Problem

Subproblem



How to Solve DFJ Formulation?

Master Problem

Subproblem

Proof of finite convergence

Proof of correctness



Group Practice 1: Longest Path Problem

In a directed graph, a team must travel from a starting location s to a destination t while visiting each location at most once and maximizing the total travel distance.



Problem 4: Facility Location



Facility Location Problem Setting

A company must decide which facilities to open and how to serve a set of customers from the open facilities while minimizing the total facility-opening and transportation costs.



Facility Location Formulation as A Two-Player Problem

A company must decide which facilities to open and how to serve a set of customers from the open facilities while minimizing the total facility-opening and transportation costs.



Problem 5: Newsvendor



Newsvendor (Expectation)

At the beginning of each week, the store must decide how many units of supplies to purchase at a unit cost c to meet the uncertain weekly demand D . Historical data provides a probability distribution for D . Any leftover inventory incurs a holding cost h per unit, while any unmet demand incurs a backlog cost b per unit. The objective is to determine the order quantity that minimizes the expected total cost.



Newsvendor (Robust 1)

At the beginning of each week, the store must decide how many units of supplies to purchase at a unit cost c to meet the uncertain weekly demand D . Historical data provides a probability distribution for D . Any leftover inventory incurs a holding cost h per unit, while any unmet demand incurs a backlog cost b per unit. The objective is to determine the order quantity that minimizes the expected total cost.



News vendor (Robust 2)

At the beginning of each week, the store must decide how many units of supplies to purchase at a unit cost c to meet the uncertain weekly demand D . Historical data provides a probability distribution for D . Any leftover inventory incurs a holding cost h per unit, while any unmet demand incurs a backlog cost b per unit. The objective is to determine the order quantity that minimizes the expected total cost.



Decision-Making under Uncertainty 1



Basic Concepts in Probability 1

- Probability $(\Omega, \mathcal{F}, \mathbb{P})$
- $\mathcal{F}$ is a σ -algebra
- $\mathbb{P}$ is the probability measure



Probability Examples

- Three-faced die

- $U[0,1]$

** The reason we cannot consider all subsets of $[0,1]$ is due to non-measurable set such as the infamous Vitali set.



Probability Visualization (as a density)

- Finite support case
- Weighed sum
- Infinite support case
- Weighted integration



Probability Visualization (as a vector)

- Finite support case
- Infinite support case



The Space of All Probability Measures

- Finite support case
- Infinite support case



Random Variables

- Random variable (as a function)
- Random variable (as a distribution)



RV Example 1

- Gambling over 3-faced die



RV Example 2

- $f(x) = x^2$ where $x \sim U[0,1]$



Expectation as Integration / Dual Paring

- Expectation
- Probability
- Moments
- Variance



Decision-Making under Uncertainty 2

- What is a decision?
- How to compare?
 - Perspective 1: aggregate random variables into a scalar (risk measures)
 - Perspective 2: change the underlying probability measures (DRO)



Review the Two Newsvendor Problems

- Expectation and worst case as different probability measures
- What is the transition between them?



Value-at-Risk (CVaR)

- Value-at-Risk β
- Example of 3-faced die
- Example of normal distribution



Conditional Value-at-Risk (CVaR)

- CVaR β
- Example of normal distribution
- Example of 3-faced die



CVaR as Robust Optimization

- The dual representation of CVaR
- Intuition



News vendor (CVaR)

At the beginning of each week, the store must decide how many units of supplies to purchase at a unit cost c to meet the uncertain weekly demand D . Historical data provides a probability distribution for D . Any leftover inventory incurs a holding cost h per unit, while any unmet demand incurs a backlog cost b per unit. The objective is to determine the order quantity that minimizes the expected total cost.



News vendor (Chance Constraints)

At the beginning of each week, the store must decide how many units of supplies to purchase at a unit cost c to meet the uncertain weekly demand D . Historical data provides a probability distribution for D . Any leftover inventory incurs a holding cost h per unit, while any unmet demand incurs a backlog cost b per unit. The objective is to determine the order quantity that minimizes the expected total cost.



Problem 6: Stochastic Shortest Path



Problem List

Later in the convex case: Portfolio: convex, multi-objective

Newsvendor / inventory: stochastic, expectation optimization

Robust shortest path optimization problem (worst-case, semi-infinite)

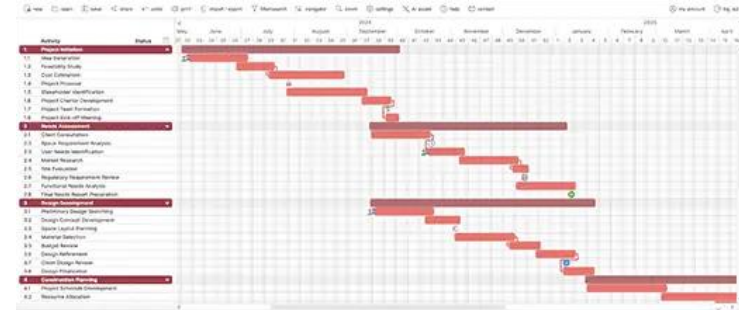
Multi-period inventory (MDP)

Bilevel pricing (two players sequential decision)



Project Management

A project has n interdependent tasks with a certain precedence relationship $i < j$ for consecutive tasks $i, j \in [n]$. Each job has a finish time t_i . How to minimize the total project time?



What if t_i is random?

